JAMES PATTERSON

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3 years of experience building and deploying ML models in production environments. I've spent my career developing computer vision systems and recommendation algorithms, working across the full pipeline from data preprocessing to model serving. I'm particularly interested in optimizing model performance at scale and translating business problems into effective technical solutions. Currently based in Oxford and looking to expand my experience with more complex deep learning architectures.

TECHNICAL SKILLS

Machine Learning: Python, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM

Data Processing: pandas, NumPy, SQL, Apache Spark, data preprocessing pipelines

Computer Vision: OpenCV, image classification, object detection, CNN architectures

Development: Python, Git, Linux, Docker, REST APIs, Jupyter notebooks

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform

Tools and Frameworks: Keras, Hugging Face, MLflow, Weights and Biases

Databases: PostgreSQL, MongoDB, Redis

WORK EXPERIENCE

Machine Learning Engineer

DataViz Solutions, London

www.dataviz.co

July 2022 - Present

Developed and maintained 5 production ML models serving real-time predictions for client analytics dashboards, achieving 94% accuracy on regression tasks and reducing inference latency by 40% through model quantization.

Built an automated retraining pipeline using Apache Airflow that monitors model drift and triggers updates when performance metrics fall below thresholds, reducing manual intervention by 60%.

Collaborated with product teams to translate business requirements into model specifications, including designing features for a customer churn prediction system that improved client retention by 12%.

Conducted code reviews for junior engineers and documented best practices for model evaluation, hyperparameter tuning, and deployment procedures across the ML team.

Machine Learning Engineer

StreamTech AI, Cambridge

www.streamtech.io

February 2021 - June 2022

Designed and implemented a computer vision pipeline for automated quality control in manufacturing, training YOLOv5 models that achieved 96% detection accuracy on defects and reduced manual inspection time by 75%.

Created feature engineering pipelines using pandas and SQL to process 2TB of daily sensor data, generating 150+ new features for predictive maintenance models.

Optimized model inference speed from 800ms to 120ms per image by implementing model distillation techniques and deploying on edge devices using TensorFlow Lite.

Established MLflow experiment tracking system for the team, standardizing hyperparameter tuning workflows and improving reproducibility across 8 ML projects.

Junior Machine Learning Engineer

Nexus Analytics, Oxford

www.nexusanalytics.net

September 2019 - January 2021

Assisted in developing recommendation algorithms for e-commerce platform, working with collaborative filtering and content-based approaches that increased click-through rates by 18%.

Performed data cleaning and exploratory analysis on customer datasets, identifying data quality issues and implementing validation checks that prevented 15 data pipeline failures.

Trained and evaluated multiple classification models (logistic regression, random forests, gradient boosting) for predicting customer lifetime value, comparing approaches and documenting findings in technical reports.

Supported deployment of ML models to AWS SageMaker, including endpoint configuration, monitoring, and A/B testing against baseline models.

EDUCATION

University of Manchester

BSc Computer Science with AI Specialisation

Graduated: June 2019

Grade: 2:1 (Upper Second-Class Honours)

GITHUB REPOSITORIES

github.com/jpatterson92/cv-defect-detection - Computer vision pipeline for manufacturing defect detection using YOLOv5, including training scripts and data annotation tools. Python, PyTorch.

github.com/jpatterson92/ml-pipeline-orchestration - Apache Airflow workflows for automated model retraining with performance monitoring and alerting. Python, PostgreSQL.

github.com/jpatterson92/time-series-forecasting - LSTM and Prophet implementations for demand forecasting with hyperparameter optimization. Python, TensorFlow.

github.com/jpatterson92/feature-engineering-toolkit - Custom pandas utilities for feature transformation, encoding, and selection with documentation and examples. Python.